

# Navigators IDR — Detailed Project Guide

## 1. Introduction

This guide explains the exact components and data flow of the Navigators IDR system.

## 2. Component Walkthrough

### 2.1 Preprocessing and Alignment

Raw IMU data is noisy and arbitrary in orientation. The system first estimates gravity to establish the *Down* vector. Then, by observing acceleration during movement, it establishes the *Forward* vector. This allows the system to rotate all raw IMU data into the vehicle frame.

### 2.2 Deep Learning (TCN) Velocity Estimation

Instead of raw double integration (which drifts quadratically in seconds), the system windows 200 timesteps (2 seconds) of aligned IMU data and passes it to an ONNX-exported Temporal Convolutional Network. The network outputs a `[v_forward, v_lateral]` prediction.

### 2.3 The 15-State EKF

The EKF state vector includes: - Position (3D) - Velocity (3D) - Attitude (3D Quaternions/Euler) - Accel Bias (3D) - Gyro Bias (3D)

The EKF prediction step uses raw IMU. The update step fuses the TCN velocity prediction (as a pseudo-measurement), NHC, ZUPT, and GNSS (when available).

## 3. Running the Validation Suite

The `scripts/benchmark.py` script runs the entire system through synthetic scenarios. It verifies that the AI model is actually running (by loading the `.pt` or `.onnx` models) and ensuring zero ground truth leaks into the estimator.